InterviewSense

Complete Technical Explanation of the Laptop-Optimized Speech Quality Pipeline

Primary method: Handcrafted acoustic features + XGBoost regression
External comparison: eGeMAPS + XGBoost
Target: NISQA Mean Opinion Score (MOS), 1–5
Evaluation: Hold-out test metrics + 5-fold GroupKFold

This document explains the project from the audio input stage through feature extraction, NISQA label repair, supervised learning, external feature comparison, cross-validation, interpretation, limitations, and what the results do—and do not—prove.

Prepared from the final InterviewSense laptop notebook and the completed experiment outputs.

1. Executive Summary

InterviewSense is a speech-analysis research pipeline designed to convert spoken audio into quantitative acoustic descriptors and use those descriptors to predict a human-rated speech quality score. The current laptop experiment is deliberately smaller than a full-scale research pipeline: it processes up to 10,000 VCTK recordings and 2,000 NISQA recordings and reuses saved feature checkpoints so that repeated notebook runs are fast.

The central supervised target is the NISQA Mean Opinion Score (MOS). NISQA provides human ratings of perceived speech quality on a 1–5 scale. Therefore, the experimentally validated task is **speech-quality regression**. It is not yet a validated interview-fluency, communication-skill, or candidate-performance score.

The project has two feature representations. The primary representation is a compact handcrafted acoustic representation consisting of 48 numeric features covering timing, silence and pauses, energy, zero-crossing rate, pitch, spectral characteristics, and 13 MFCCs with mean and standard deviation statistics. The comparison representation is eGeMAPSv02, an established speech-feature set. Both representations are evaluated using the same XGBoost regression family so that the main experimental difference is the feature representation.

The completed run produced 12,000 extracted feature rows: 10,000 VCTK and 2,000 NISQA. Correct NISQA metadata matching recovered 1,036 valid MOS labels from the 2,000 NISQA feature rows. The primary model achieved a test MAE of 0.5912, RMSE of 0.7240, and R^2 of 0.4797. The eGeMAPS comparison achieved MAE 0.5731, RMSE 0.7301, and R^2 0.4709.

What the project demonstrates

- How raw speech can be transformed into structured acoustic measurements.
- How a human-rated speech-quality target can be recovered from corpus metadata.
- How a regression model can learn a relationship between acoustic characteristics and perceived speech quality.
- How an established feature representation such as eGeMAPS can be used as an external baseline.
- How repeated evaluation through 5-fold GroupKFold can provide a more stable estimate than a single split.
- How to build the pipeline so expensive audio feature extraction is checkpointed and does not need to be repeated.

Most important scope statement

Do not present the current MOS predictor as an interview communication score. The model has been trained and evaluated against NISQA MOS. To make a scientifically valid claim about interview fluency or communication quality, the project needs a human-rated interview/communication dataset whose labels directly represent that target.

2. Problem Statement and Motivation

Speech contains information at several levels. Some information is linguistic—what a person says—while other information is paralinguistic or acoustic—how the speech sounds. Timing, pauses, energy, pitch, spectral distribution and cepstral characteristics can all be measured without requiring the system to understand the entire semantic content of an utterance.

InterviewSense uses this acoustic route because it is computationally practical on a laptop and provides interpretable measurements. The idea is to create a repeatable numerical representation of speech and then learn a mapping from that representation to a human rating.

The research question in the present experiment can therefore be stated as:

Can a compact handcrafted acoustic representation predict human-rated speech quality, and how does its performance compare with an established speech representation such as eGeMAPS?

Why this is useful for the larger InterviewSense idea

- A speech-analysis system needs measurable signals before it can produce feedback.
- Acoustic features are inexpensive compared with large end-to-end deep-learning systems.
- A compact feature vector makes it easier to inspect which properties influence predictions.
- An external baseline prevents the project from evaluating its own representation in isolation.
- NISQA provides a genuine human-rated numeric target, making it suitable for a supervised regression baseline.

3. Overall System Architecture

The complete pipeline can be understood as the following sequence:

Raw Audio → Dataset Sampling → Audio Normalization → Acoustic Feature Extraction → Feature Checkpointing → NISQA MOS Label Mapping → Labelled Dataset → Group-Based Split → XGBoost Regression → Test Evaluation → eGeMAPS Baseline → 5-Fold Group Cross-Validation → Feature Importance + Diagnostics

1. **Dataset acquisition:** VCTK and NISQA are obtained through KaggleHub.
2. **Sampling:** only 10,000 VCTK and 2,000 NISQA recordings are used in the laptop experiment.
3. **Audio processing:** audio is loaded at 16 kHz and converted to mono.
4. **Feature extraction:** timing, pause, energy, pitch, spectral and MFCC descriptors are computed.
5. **Checkpointing:** feature CSVs are saved in batches so interrupted runs can resume.
6. **Label repair:** NISQA MOS is read from the correct corpus CSV and matched by audio filename.
7. **Supervised subset:** only NISQA rows with valid numeric MOS labels are used for training.
8. **Split:** the notebook uses GroupShuffleSplit for train/validation/test partitioning.
9. **Primary model:** XGBRegressor learns to predict MOS from the 48 handcrafted features.
10. **External baseline:** eGeMAPSv02 features are extracted from the same labelled recordings and passed to the same XGBoost family.
11. **Validation:** the primary representation is evaluated with 5-fold GroupKFold.
12. **Interpretation:** metrics, feature importance and true-vs-predicted diagnostics are produced.

4. Datasets

Dataset	Role in project	Amount used	Label status
VCTK SR16k	Acoustic feature development / broader speech feature population	10,000 recordings	No supervised MOS target used
NISQA	Supervised speech-quality regression	2,000 recordings processed	1,036 MOS-labelled rows used
Speech Accent Archive	Configured reference dataset	Not processed in final supervised run	No compatible target used
Santa Barbara Corpus	Configured reference dataset	Not processed in final supervised run	No compatible target used

VCTK

The VCTK SR16k Kaggle dataset is used as a large English multi-speaker speech source. The notebook samples 10,000 recordings rather than processing all available recordings. The filename pattern also allows a speaker identifier to be inferred for VCTK recordings such as p225_001.wav → p225.

NISQA

NISQA is the key supervised dataset. Its metadata contains human-rated MOS values. The metadata file used in the successful run contains 14,432 rows and 21 columns, including fields such as `filename_deg`, `filename_ref`, `mos`, noise, coloration, distortion, loudness and their standard deviations.

All 14,432 metadata rows have a non-null MOS in the metadata table. The feature pipeline, however, only processed 2,000 NISQA audio files. Filename-based matching successfully recovered MOS for 1,036 of those 2,000 processed recordings.

Why the other datasets are not supervised targets

The Speech Accent Archive and Santa Barbara Corpus are retained in configuration/reference context, but the final notebook does not use them as supervised training targets because their available metadata does not provide a directly compatible human-rated interview/fluency target.

5. Acoustic Feature Extraction

The project's main representation is handcrafted rather than a learned embedding. The extractor uses librosa to convert each waveform into a fixed-length numerical vector. Fixed-length vectors are important because XGBoost expects tabular input where each row represents one recording and each column represents a feature.

Audio preprocessing

- Audio is loaded using librosa at a target sampling rate of 16,000 Hz.
- Audio is converted to mono.
- Empty audio files are rejected.
- The waveform duration is calculated from the number of samples divided by the sampling rate.
- A simple energy-based speech/silence segmentation is obtained with librosa.effects.split.

Feature family 1 — Timing and speech activity

Feature	Meaning
duration	Total recording duration in seconds.
speech_duration	Estimated time classified as speech.
silence_duration	Duration not classified as speech.
speech_ratio	speech_duration / duration.
pause_count	Number of positive gaps between detected speech intervals.
avg_pause_duration	Average duration of detected pauses.
max_pause_duration	Longest detected pause.

These features quantify temporal organization. For example, a recording with large silence_duration or a low speech_ratio contains relatively less detected speech activity. Pause statistics describe the distribution of gaps

between speech regions.

Feature family 2 — Energy

Feature	Meaning
rms_mean	Mean root-mean-square energy across frames.
rms_std	Standard deviation of frame-level RMS energy.

RMS energy is a measure of signal amplitude. Its mean captures overall energy level while its standard deviation captures how much the energy changes over time.

Feature family 3 — Zero-crossing rate

Feature	Meaning
zcr_mean	Mean zero-crossing rate.
zcr_std	Standard deviation of zero-crossing rate.

Zero-crossing rate measures how frequently the waveform changes sign. It is sensitive to signal characteristics such as noisiness and high-frequency content and can complement spectral features.

Feature family 4 — Pitch

Feature	Meaning
pitch_mean	Mean finite fundamental-frequency estimate.
pitch_median	Median fundamental-frequency estimate.
pitch_std	Pitch variability.
pitch_min	Minimum finite pitch estimate.
pitch_max	Maximum finite pitch estimate.
pitch_range	pitch_max – pitch_min.
voiced_frame_ratio	Fraction of frames with a finite pitch estimate.

Pitch is estimated using librosa.pyin with a search range around the human voice region. The extractor removes non-finite pitch estimates before computing summary statistics.

Feature family 5 — Spectral features

Feature	Meaning
spec_centroid_mean	Average spectral centroid.
spec_centroid_std	Variation in spectral centroid.
spec_bandwidth_mean	Average spectral bandwidth.
spec_bandwidth_std	Variation in spectral bandwidth.

The spectral centroid can be interpreted as a brightness-related center of mass of the spectrum, while spectral bandwidth describes how spread out the spectrum is around its centroid.

Feature family 6 — MFCCs

The extractor computes 13 Mel-frequency cepstral coefficients. For each MFCC, the notebook stores both its mean and standard deviation. That produces 13 × 2 = 26 MFCC-derived features.

MFCCs summarize the short-term spectral envelope on a perceptually motivated Mel scale. They are widely used in speech and audio processing because they provide a compact representation of spectral shape.

Final handcrafted feature count

48 features total: 7 timing/silence + 2 RMS + 2 ZCR + 7 pitch/voicing + 4 spectral + 26 MFCC statistics.

6. Laptop Optimization and Checkpointing

A major engineering decision was to make the pipeline practical on a normal laptop. The notebook does not use Whisper in the current run and uses CPU-parallel acoustic extraction. It also saves feature checkpoints every 250 recordings.

- VCTK limit: 10,000 recordings.
- NISQA limit: 2,000 recordings.
- Whisper: disabled for the laptop run.
- Checkpoint interval: 250 files.
- Existing VCTK/NISQA feature CSVs are loaded if they already exist.
- Only unprocessed file keys are extracted on a resumed run.
- Feature rows are deduplicated by file_key.

This checkpoint design explains why later notebook executions were fast. The expensive operation is audio feature extraction. Once the feature CSVs exist, the notebook can load them directly and move to label repair and modelling without recomputing thousands of waveforms.

7. NISQA MOS Label Repair — The Critical Data Fix

The original pipeline had a major data-integration problem: the extracted NISQA rows existed, but the target column contained no usable MOS labels. The corrected notebook explicitly searches for a CSV containing both a MOS column and a file-path/filename column.

The successful metadata file was the NISQA corpus CSV. It contains the degraded filename and MOS. The notebook takes the basename of the metadata path and the basename of each extracted audio file, then creates a dictionary from filename to MOS.

For every NISQA feature row, the target is assigned by looking up the lowercased filename. This is why the repaired pipeline changed the target availability from 0 labelled NISQA rows to 1,036 labelled rows.

Successful mapping result

Quantity	Result
NISQA metadata rows	14,432
NISQA metadata columns	21
Processed NISQA feature rows	2,000
Matched MOS labels	1,036
Unmatched MOS labels	964

Quantity	Result
MOS minimum	1.000
MOS maximum	5.000
MOS mean	2.908
MOS median	2.800

Only the 1,036 rows with genuine numeric MOS values are allowed into supervised learning. VCTK remains useful for feature development but does not enter the MOS regression target set.

8. Supervised Learning Dataset

After label repair, the notebook filters df_features to rows where dataset = NISQA and target is non-null. The target is explicitly converted to numeric and any remaining invalid values are removed.

Property	Value
Supervised dataset	NISQA
Labelled samples	1,036
Target	NISQA MOS
Target range	1.0–5.0
Primary feature count	48
Task type	Regression
Model family	XGBoost Regressor

Non-feature columns such as dataset, speaker_id, file_name, file_key, target and group_id are excluded. Only numeric acoustic columns are passed to the model.

9. Train / Validation / Test Strategy

The notebook uses GroupShuffleSplit rather than a completely unrestricted random row split. The intention is to keep recordings belonging to the same group together. The first split creates approximately 70% training and 30% temporary data. The temporary data is then split equally into validation and test, giving an intended 70/15/15 structure.

Split	Samples in completed run	Approx. proportion
Train	725	70%
Validation	155	15%
Test	156	15%
Total	1,036	100%

The notebook performs explicit intersection checks between train, validation and test group sets. The completed run reported that the group leakage test passed.

Important interpretation of the current grouping

The notebook constructs NISQA group_id by removing the final condition suffix from the filename, e.g. c00001_for_cnv_f_0104_02.wav becomes c00001_for_cnv_f_0104. In the completed run, however, the reported

labelled-sample group count was effectively one group per labelled sample. Therefore, the GroupShuffleSplit mechanism itself was used correctly, but the available grouping did not provide strong speaker/source aggregation in that run.

For reporting, it is safer to say that the experiment used a group-aware split and passed its explicit non-overlap check, rather than claiming that the results are fully speaker-independent.

10. Numerical Preprocessing

Before modelling, the feature matrices are cleaned. Positive and negative infinity values are replaced with NaN. Missing values are then filled using medians calculated from the training set.

The same training medians are applied to validation and test data. This is important because computing imputation statistics from the full dataset could leak information from the validation/test sets into training.

11. Primary Model — XGBoost Regression

The primary model is XGBRegressor. XGBoost is a gradient-boosted decision-tree method. Instead of training one tree, it builds a sequence of trees where each new tree attempts to correct errors made by the previous ensemble.

For this project, the model receives a 48-dimensional acoustic vector and outputs one continuous number: the predicted NISQA MOS.

Hyperparameter	Value
n_estimators	500
max_depth	5
learning_rate	0.03
subsample	0.8
colsample_bytree	0.8
objective	reg:squarederror
eval_metric	rmse
random_state	42
n_jobs	-1

Why regression?

MOS is continuous. A score of 3.7 contains more information than simply assigning a class such as good or bad. Regression therefore preserves the numeric nature of the human rating.

12. Evaluation Metrics

Metric	Meaning	Desired direction
MAE	Average absolute prediction error in MOS units.	Lower is better
RMSE	Square root of average squared prediction error; penalizes larger errors more strongly.	Lower is better
R^2	Fraction of target variance explained relative to a constant-mean baseline.	Higher is better

MAE intuition

An MAE of 0.5912 means that, on average, the model's predicted MOS differs from the human MOS by about 0.59 MOS points on the evaluated test set.

RMSE intuition

RMSE of 0.7240 is larger than MAE because RMSE gives more weight to larger mistakes. It is therefore useful for detecting whether a model occasionally makes substantially wrong predictions.

R^2 intuition

R^2 = 0.4797 means the model explains approximately 48% of the variance in the held-out target values relative to a mean-only reference. It does not mean the model is 47.97% accurate.

13. Primary Model Results

Metric	Primary handcrafted features + XGBoost
MAE	0.5912
RMSE	0.7240
R^2	0.4797

These results indicate that the acoustic representation contains substantial predictive information about perceived speech quality. At the same time, the R^2 value shows that a large portion of variation remains unexplained. That is expected for a compact acoustic representation predicting subjective human ratings.

14. External Validation Model — eGeMAPS

The faculty requirement for comparison with an existing approach is addressed through eGeMAPS. eGeMAPS is an established speech feature representation designed to capture interpretable voice and paralinguistic characteristics. The project uses eGeMAPSv02 at the Functionals level.

The comparison is intentionally controlled: the same labelled recordings are used, the same target is used, and the same XGBoost regression family is used. Therefore, the feature representation is the major change between the two experiments.

Approach	Feature count	MAE	RMSE	R^2
InterviewSense handcrafted features + XGBoost	48	0.5912	0.7240	0.4797
eGeMAPS + XGBoost	88	0.5731	0.7301	0.4709

Interpretation: eGeMAPS obtains a slightly lower MAE, while the InterviewSense handcrafted representation obtains slightly lower RMSE and slightly higher R^2. Therefore, the two approaches are best described as broadly comparable rather than declaring an outright winner.

Why this is a good cross-validation comparison

- It compares against an established speech representation rather than an arbitrary second classifier.
- Both representations are evaluated on the same task.
- Both use XGBoost, reducing the chance that the comparison is mainly caused by a different model family.

- MAE, RMSE and R^2 provide complementary views of performance.
- The comparison shows that a compact 48-feature representation can remain competitive with a larger established representation.

15. Five-Fold Group Cross-Validation

A single train/test split can be sensitive to which recordings happen to fall into each partition. The notebook therefore performs five-fold GroupKFold evaluation on the primary representation.

For each fold, the data is divided into training and validation subsets according to group IDs. An XGBoost model is trained from scratch for that fold and evaluated on its held-out fold. The notebook records MAE, RMSE and R^2 for every fold.

Fold	MAE	RMSE	R^2
1	0.6648	0.8588	0.3765
2	0.6380	0.7890	0.5014
3	0.6405	0.8291	0.4138
4	0.6599	0.8031	0.4440
5	0.6238	0.7927	0.4598
Mean ± SD	0.6454 ± 0.0168	0.8145 ± 0.0293	0.4391 ± 0.0472

The standard deviations quantify how much the metric changes from fold to fold. The relatively modest MAE variation indicates reasonably consistent error, while R^2 varies more because explained variance is sensitive to the distribution of target values in each fold.

16. Feature Importance and Interpretability

The notebook extracts XGBoost feature_importances_ values and sorts the 48 handcrafted features from most to least important. This creates a feature-importance table that can be saved as primary_feature_importance.csv.

Feature importance should be interpreted as model usage, not as proof of causation. If a feature has high importance, it means the fitted model relied strongly on it when making predictions; it does not prove that changing that acoustic property would necessarily change the human MOS.

The feature families themselves provide a useful interpretability layer: timing features describe speech/silence organization, pitch features describe fundamental-frequency behavior, energy features describe amplitude, spectral features describe frequency distribution, and MFCC statistics describe spectral-envelope characteristics.

17. Prediction Diagnostics

The notebook also produces a true-versus-predicted MOS scatter plot. The x-axis contains the true NISQA MOS values and the y-axis contains model predictions. A diagonal reference line represents perfect prediction.

Points close to the diagonal indicate good predictions. Systematic compression toward the center can reveal a model that predicts extreme ratings conservatively. Outliers far from the diagonal identify recordings where acoustic information was insufficient or the learned relationship did not generalize well.

18. Output Files and What They Mean

File	Purpose
features_all_repaired.csv	12,000-row processed feature dataset with repaired NISQA targets where available.
features_all.csv	Saved feature dataset from the notebook's label-repair stage.
model_comparison_results.csv	Primary handcrafted representation versus eGeMAPS comparison metrics.
group_cross_validation_results.csv	Per-fold 5-fold GroupKFold metrics.
primary_feature_importance.csv	Feature-importance ranking from the primary XGBoost model.
feature_checkpoints/vctk_features.csv	Checkpointed VCTK acoustic features.
feature_checkpoints/nisqa_features.csv	Checkpointed NISQA acoustic features.

The checkpoint files are particularly important for laptop use. They prevent repeated audio extraction. The final CSV outputs are useful for reports, plots, and reproducibility.

19. End-to-End Walk-Through of One Audio File

1. An audio file is selected from VCTK or NISQA.
2. librosa loads it at 16 kHz and converts it to mono.
3. The waveform duration is calculated.
4. RMS and zero-crossing-rate sequences are calculated frame by frame.
5. Speech intervals are detected using energy-based splitting.
6. Speech duration, silence duration, speech ratio and pause statistics are derived.
7. pYIN estimates pitch; finite pitch values are summarized.
8. An STFT is computed and spectral centroid/bandwidth statistics are obtained.
9. 13 MFCCs are calculated and each coefficient is summarized by mean and standard deviation.
10. The resulting 48-value vector is stored together with dataset, speaker/file identifiers and target placeholder.
11. The feature row is checkpointed to CSV.
12. For NISQA, the filename is matched against the NISQA metadata and MOS is attached.
13. Only rows with valid MOS become supervised examples.
14. The example is assigned to a group and placed into the group-aware train/validation/test workflow.
15. XGBoost uses the acoustic vector to predict MOS.
16. The same recording can also be represented by eGeMAPS for the external comparison.

20. Why VCTK Is Not the Training Target

A dataset can be useful for feature engineering without being suitable as the supervised target dataset. VCTK provides a large amount of multi-speaker speech, but the final experiment needs a numeric human-rated target. NISQA provides that target through MOS.

Combining VCTK and NISQA directly into one supervised target without a common label definition would be methodologically incorrect. Therefore, VCTK is retained as a feature-development resource while NISQA supplies

the labelled regression examples.

21. Why the Laptop Uses 10,000 + 2,000 Instead of the Full Corpora

The project is engineered for a laptop rather than a GPU workstation. Processing every available audio recording would increase extraction time and storage requirements without being necessary to demonstrate the core experimental pipeline.

- 10,000 VCTK recordings provide a substantial acoustic feature-development population.
- 2,000 NISQA recordings are enough to create a useful supervised subset while keeping extraction practical.
- Checkpointing allows the experiment to resume instead of starting over.
- The supervised experiment ultimately uses only the 1,036 NISQA rows for which MOS was successfully matched.

This is a reasonable prototype-scale experimental design. A final research paper could scale the sample size after the pipeline has been validated.

22. What the Model Is Actually Learning

The model is not directly learning the meaning of an interview answer. It is learning statistical relationships between measurable acoustic patterns and perceived speech quality ratings in the NISQA corpus.

For example, a combination of timing, energy, pitch and spectral characteristics may correlate with higher or lower perceived quality. XGBoost can represent nonlinear interactions among these variables. It does not require a linear assumption such as 'more pitch variation always means better speech.'

Because the target is subjective MOS, there will naturally be irreducible variation. Different listeners can rate the same speech differently, and acoustic features do not capture every factor that influences perception.

23. Strengths of the Current Project

- Clear separation between feature extraction and supervised target construction.
- A genuine human-rated numeric target rather than a synthetic label.
- A compact and interpretable 48-feature representation.
- Laptop-friendly implementation with checkpointing and CPU parallelism.
- External comparison against established eGeMAPS features.
- Multiple evaluation metrics rather than relying on one number.
- Five-fold cross-validation in addition to a held-out test set.
- Saved CSV outputs make the experiment easy to inspect and report.
- The project explicitly documents that NISQA MOS is speech quality rather than interview communication.

24. Limitations

Limitation	Consequence
NISQA MOS measures speech quality, not interview communication	Current model cannot be claimed as a validated interviewer/candidate communication score.

Limitation	Consequence
Only 1,036 of 2,000 processed NISQA rows received MOS	Supervised sample size is smaller than the processed audio count.
Current group IDs are not strong speaker-level groups in the completed run	The group-aware mechanism passed non-overlap checks, but speaker-independent generalization is not fully established.
Handcrafted features are compact	Some linguistic, contextual and semantic information is absent.
XGBoost is a tabular regressor	It does not directly model raw waveform sequences or long temporal dependencies.
Subjective MOS contains listener variability	Perfect prediction is not expected.
Laptop-scale sampling	Results may change with a larger and more diverse sample.

25. Future Work — Turning the Baseline into InterviewSense

The current experiment is best viewed as the validated acoustic foundation of InterviewSense. The next stage should change the target from generic speech quality to the actual interview objective.

1. Collect or obtain a human-rated interview dataset with labels such as fluency, clarity, confidence, relevance, articulation or communication quality.
2. Define a precise target and annotation protocol before training.
3. Use the existing 48-feature extractor as a baseline representation.
4. Compare the handcrafted features with eGeMAPS and, later, learned speech embeddings.
5. Use verified speaker-level grouping so that a speaker cannot appear across training and evaluation.
6. Evaluate generalization on completely unseen speakers.
7. Add temporal models if frame-level or segment-level information becomes necessary.
8. Build a feedback layer that maps model outputs to actionable recommendations.
9. Add calibration and uncertainty estimation so the system can indicate when a prediction is unreliable.
10. Perform ablation studies to determine which feature families actually contribute to performance.

26. Proposed Final InterviewSense Architecture

Microphone / Audio File
↓
Audio Preprocessing — resampling, mono conversion, quality checks
↓
Speech Segmentation / VAD — speech, silence and pauses
↓
Acoustic Feature Extraction — timing + energy + pitch + spectrum + MFCC/eGeMAPS
↓
Speaker-Level Aggregation
↓
Validated Interview Model — regression or multi-task prediction
↓
Score + Confidence
↓

Feedback Engine — specific, evidence-based interview recommendations

27. Faculty / Viva Questions You Should Be Ready For

What is the exact target?

NISQA MOS, a human-rated perceived speech-quality score from 1 to 5.

Is this an interview communication score?

No. The current experiment validates speech-quality regression. A human-rated interview dataset is required for that claim.

Why XGBoost?

The input is structured tabular acoustic features, and XGBoost can model nonlinear relationships efficiently on CPU.

Why not classify MOS?

MOS is naturally continuous, so regression preserves more information than forcing it into discrete classes.

Why use eGeMAPS?

It is an established speech-feature representation and provides an external feature-level baseline.

Why use the same XGBoost model for comparison?

It keeps the regressor approximately constant so the feature representation is the main experimental difference.

Why do you need cross-validation?

A single split can be noisy. Five-fold GroupKFold shows how performance varies across several held-out folds.

What is MAE?

Mean absolute error: the average absolute difference between predicted and true MOS.

What does $R^2 = 0.4797$ mean?

The model explains about 48% of the variance in the evaluated target relative to a mean-only baseline; it is not 48% accuracy.

Why did the notebook become fast after the first run?

The expensive acoustic features were saved as checkpoints and loaded instead of recomputed.

Why only 1,036 labelled samples?

Only those NISQA recordings had a successful filename-to-MOS metadata match among the 2,000 processed recordings.

Why not use all VCTK for training?

VCTK does not provide the same compatible human-rated MOS target used by NISQA.

28. How to Present the Results

A strong presentation should not claim that the model is perfect or that it already measures interview skill. Instead, frame the work as a controlled acoustic speech-quality experiment and explain how it forms the technical foundation for a future interview-specific model.

Recommended statement: “We developed a laptop-optimized acoustic speech-quality regression pipeline using 48 handcrafted acoustic features and XGBoost. On the current NISQA experiment, the model achieved MAE 0.5912, RMSE 0.7240 and R^2 0.4797. We compared the representation against eGeMAPSv02 using the same XGBoost regression family, where eGeMAPS achieved MAE 0.5731, RMSE 0.7301 and R^2 0.4709. Five-fold GroupKFold on the primary representation produced mean MAE 0.6454 ± 0.0168, RMSE 0.8145 ± 0.0293 and R^2 0.4391 ± 0.0472.”

29. Reproducibility Checklist

- Use the interviewsense Python environment.
- Ensure numpy, pandas, scipy, scikit-learn, librosa, soundfile, tqdm, joblib, kagglehub, xgboost, opensmile and plotting dependencies are installed.
- Keep the random seed at 42 for the reported experiment.
- Keep the 10,000 VCTK / 2,000 NISQA laptop limits if reproducing the current run.
- Keep the feature checkpoint directory so extraction does not restart.
- Use the NISQA corpus CSV containing MOS and filepath information.
- Confirm that approximately 1,036 NISQA rows receive MOS before training.
- Run the same XGBoost hyperparameters.
- Run both the primary and eGeMAPS comparisons.
- Save the three principal result CSVs.

30. Conclusion

InterviewSense currently provides a complete, reproducible acoustic speech-quality baseline. The system starts with speech recordings, extracts a compact 48-dimensional acoustic representation, repairs genuine NISQA MOS labels, trains an XGBoost regression model, compares that representation against eGeMAPS, and evaluates stability through five-fold group-aware cross-validation.

The completed experiment demonstrates that the handcrafted representation contains useful information for predicting perceived speech quality. Its test performance is competitive with the eGeMAPS comparison under the same XGBoost modelling family: the handcrafted representation has slightly better RMSE and R^2, while eGeMAPS has slightly better MAE.

The central scientific limitation is also clear: NISQA MOS is a speech-quality label, not an interview communication label. Consequently, the current project should be presented as the acoustic baseline and methodological foundation for InterviewSense. The next major research step is to obtain human-rated interview data and retrain/evaluate the system against the actual communication objectives.

Bottom line: the current pipeline is not a fake or placeholder experiment. It has a real human-rated target, a reproducible feature-extraction process, a primary regression model, an established external feature comparison, and cross-validation. What remains is to connect that validated acoustic machinery to a genuinely interview-specific target.

31. Source and Scope Note

This document was prepared from the uploaded/final InterviewSense_Laptop_Corrected_CrossValidation notebook, the project feature-extraction code, and the completed experiment outputs reported during the project work. The

notebook itself defines the pipeline as a laptop-optimized speech-quality experiment and explicitly warns against interpreting NISQA MOS as an interview communication score.

The exact model results, split sizes and NISQA mapping statistics in this report correspond to the completed run discussed during the project session. If the notebook is rerun with different samples, package versions, or changed random seeds, small metric differences are possible.